

Physical Chemistry of Materials

Lecture 05

Applications of Canonical Ensemble



Last Class

(Thermodynamic) Ensemble

Idealized group of a large number of virtual copies
(sometimes infinitely many) of a system

Microcanonical ensemble

Canonical ensemble

Grand canonical ensemble

⋮



Microcanonical Ensemble

Thermodynamic ensemble of an **isolated system**

- Conserved quantities: N, V, U (or traditionally N, V, E)
- All microstates are equally probable
- Probability to find the system in microstate j

$$p_j = \frac{1}{W}$$

Canonical Ensemble

Thermodynamic ensemble of a **closed system**

- Conserved quantities: N, V, T
- Microstates may have different energies
- All microstates are NOT equally probable
- Probability to find the system in some microstate?

Last Class

Principle of Maximum Entropy

The probability distribution which **best represents**
the current state of knowledge about a system is
the one with largest entropy



Ensemble Energy Conservation

$$S/k_B = -\sum_i p_i \ln p_i$$

$$\sum_i p_i - 1 = 0$$

We can introduce one more constraint: $\sum_i p_i E_i = U$

As the constraint implies $\sum_i p_i E_i - U = 0$,

$$\frac{\partial [S/k_B - \lambda(\sum_i p_i - 1) - \beta(\sum_i p_i E_i - U)]}{\partial p_i} = 0$$

$$p_i = e^{-1-\lambda-\beta E_i}$$

Summary

For a canonical ensemble, the probability to find the system in a microstate i (with energy E_i) is

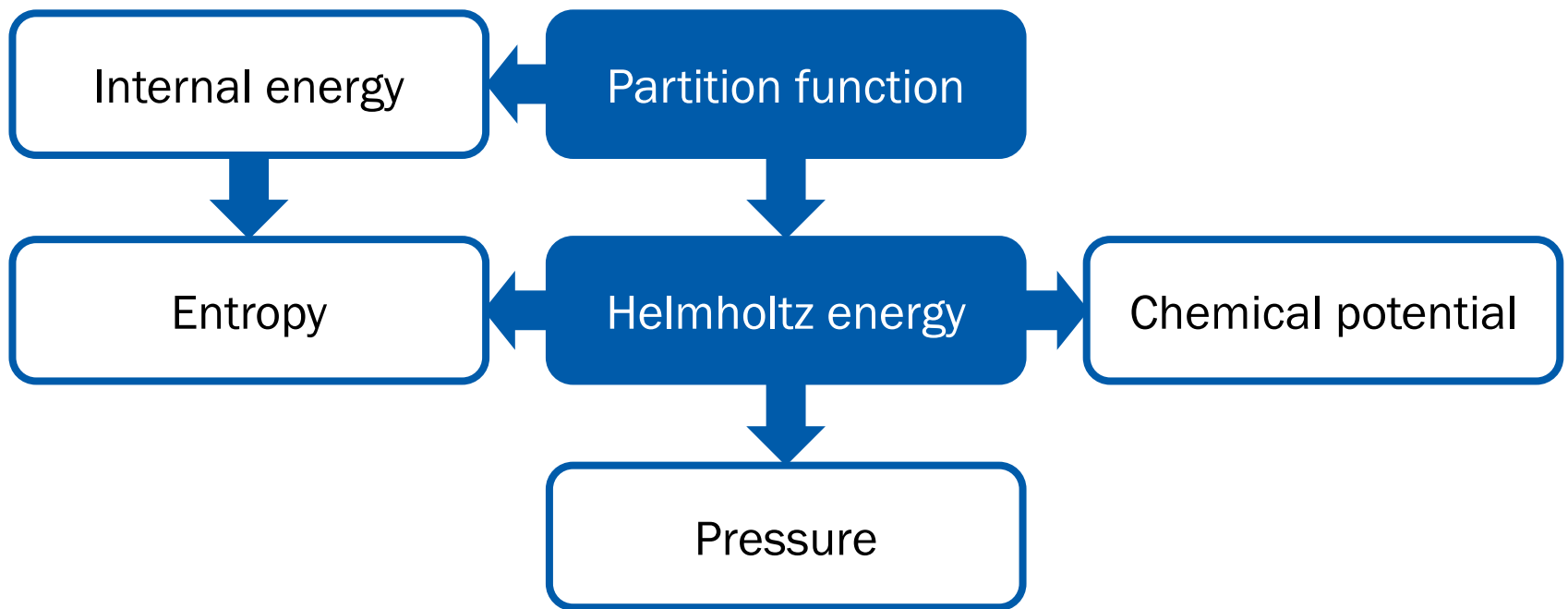
$$p_i = e^{-1-\lambda-\beta E_i} = \frac{e^{-\beta E_i}}{Z} = \frac{e^{-E_i/k_B T}}{Z}$$

where the partition function $Z = \sum_i e^{-\beta E_i} = \sum_i e^{-E_i/k_B T}$

Also, the Helmholtz energy is given by

$$A = -k_B T \ln Z$$

Z and Thermodynamics



Multi-Particle Partition Functions

- Independent & distinguishable particles

$$Z = z^N \text{ (} z: \text{ one-particle partition function)}$$

- Independent & indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

- Interacting particles: later

Two Distinguishable Particles

- Consider two particles A and B, with energies ε_i^A and ε_j^B
- Partition function of A: $z_A = \sum_i e^{-\beta \varepsilon_i^A}$
- Partition function of B: $z_B = \sum_j e^{-\beta \varepsilon_j^B}$
- Total energy: $E_{ij} = \varepsilon_i^A + \varepsilon_j^B$
- Total partition function

$$Z = \sum_{i,j} e^{-\beta E_{ij}} = \sum_i \sum_j e^{-\beta(\varepsilon_i^A + \varepsilon_j^B)} = z_A z_B$$

N Distinguishable Particles

Similarly,

$$Z = z_A z_B \cdots z_X$$

If the one-particle partition functions are identical ($= z$),

$$Z = z^N$$

Two Indistinguishable Particles

For two indistinguishable particles,

particle 1 in state A & particle 2 in state B

= particle 1 in state B & particle 2 in state A

Hence, the “correct” partition function is

$$Z = \frac{Z_A Z_B}{2}$$

N Indistinguishable Particles

We need to consider Gibbs' correction factor: $1/N!$

$$Z = \frac{1}{N!} z_A z_B \cdots z_X$$

If the one-particle partition functions are identical ($= z$),

$$Z = \frac{1}{N!} z^N$$

Two-State Example

Consider a particle with an internal state: state A or state B.

$$E(A) = \epsilon$$

$$E(B) = -\epsilon$$

The one-particle partition function is

$$z = \sum_s e^{-\beta E(s)} = e^{-\beta E(A)} + e^{-\beta E(B)}$$
$$z = e^{-\beta\epsilon} + e^{+\beta\epsilon} = 2 \cosh(\beta\epsilon)$$

Partition Function

$$z = e^{-\beta\epsilon} + e^{+\beta\epsilon} = 2 \cosh(\beta\epsilon)$$

- Distinguishable particles:

$$Z = z^N = [2 \cosh(\beta\epsilon)]^N$$

- Indistinguishable particles:

$$Z = \frac{1}{N!} z^N = \frac{1}{N!} [2 \cosh(\beta\epsilon)]^N$$

Thermodynamics

- Helmholtz energy

$$A = -k_B T \ln Z = -N k_B T \ln[2 \cosh(\beta \epsilon)]$$

- Internal energy

$$U = -\frac{\partial \ln Z}{\partial \beta} = -N \frac{\partial}{\partial \beta} \ln[2 \cosh(\beta \epsilon)]$$

$$\therefore U = -N \epsilon \tanh(\beta \epsilon)$$

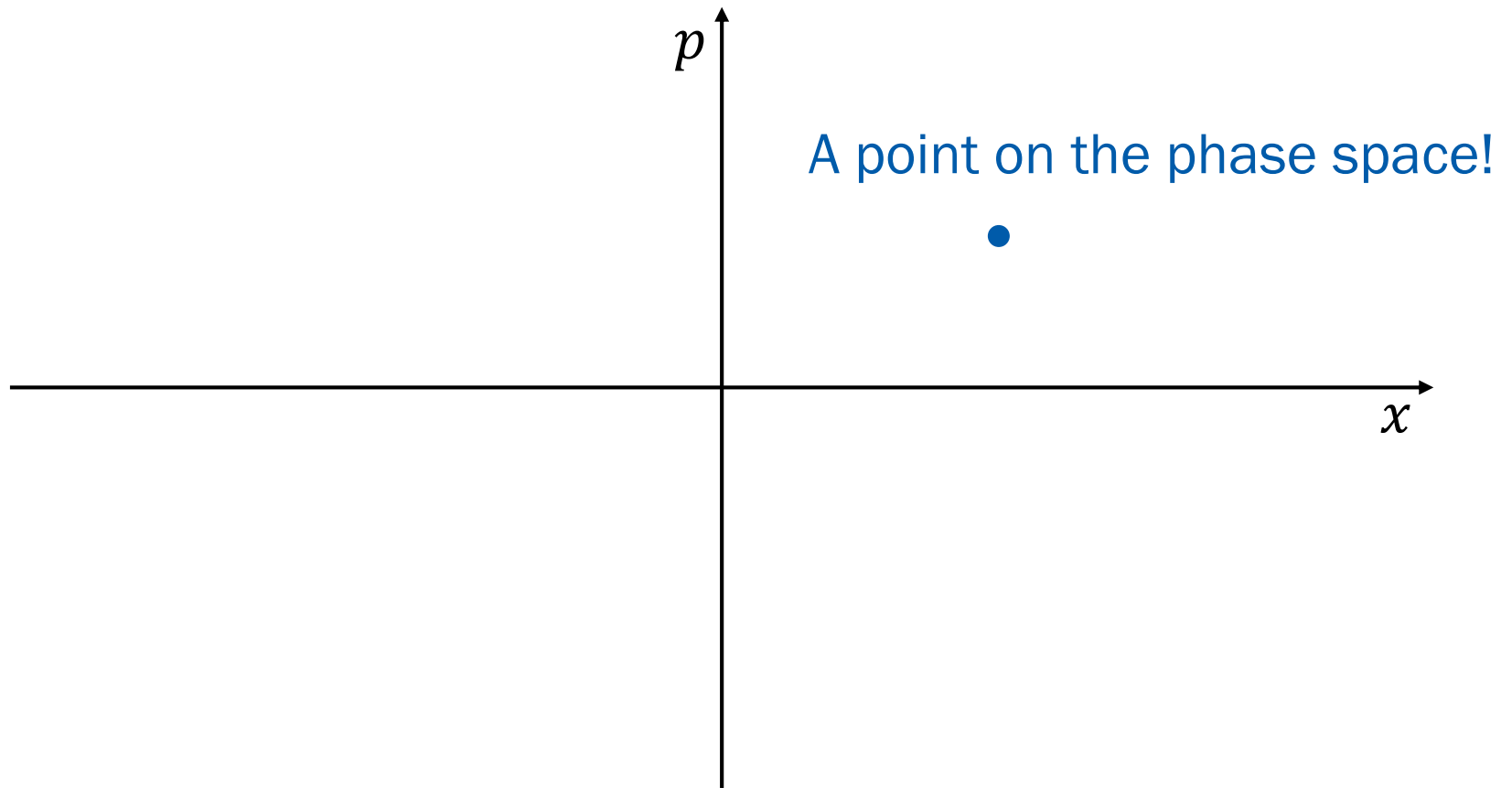
- Entropy: $S = (U - A)/T$

$$S = \frac{1}{T} \{-N \epsilon \tanh(\beta \epsilon) + N k_B T \ln[2 \cosh(\beta \epsilon)]\}$$

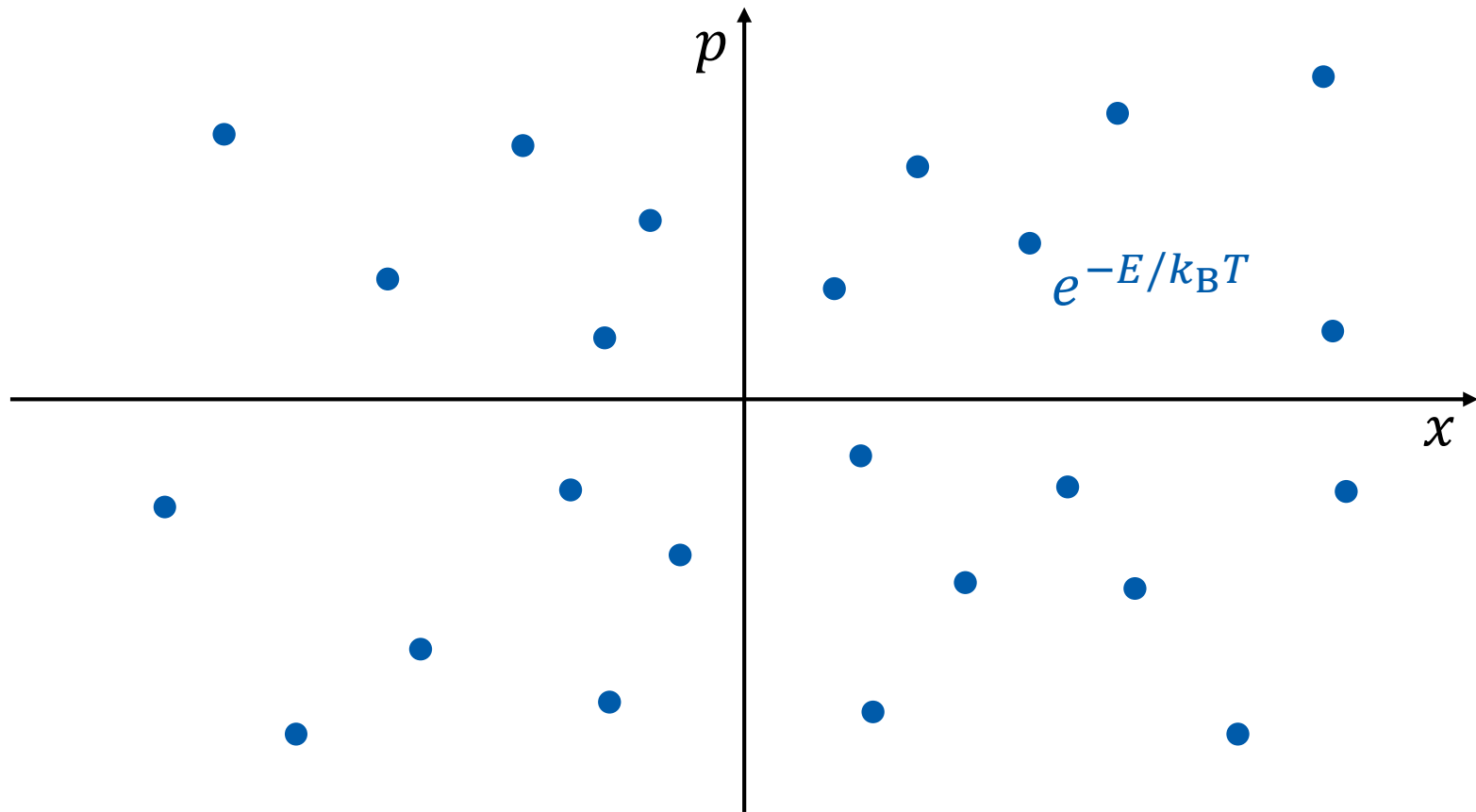


Classical Statistical Mechanics

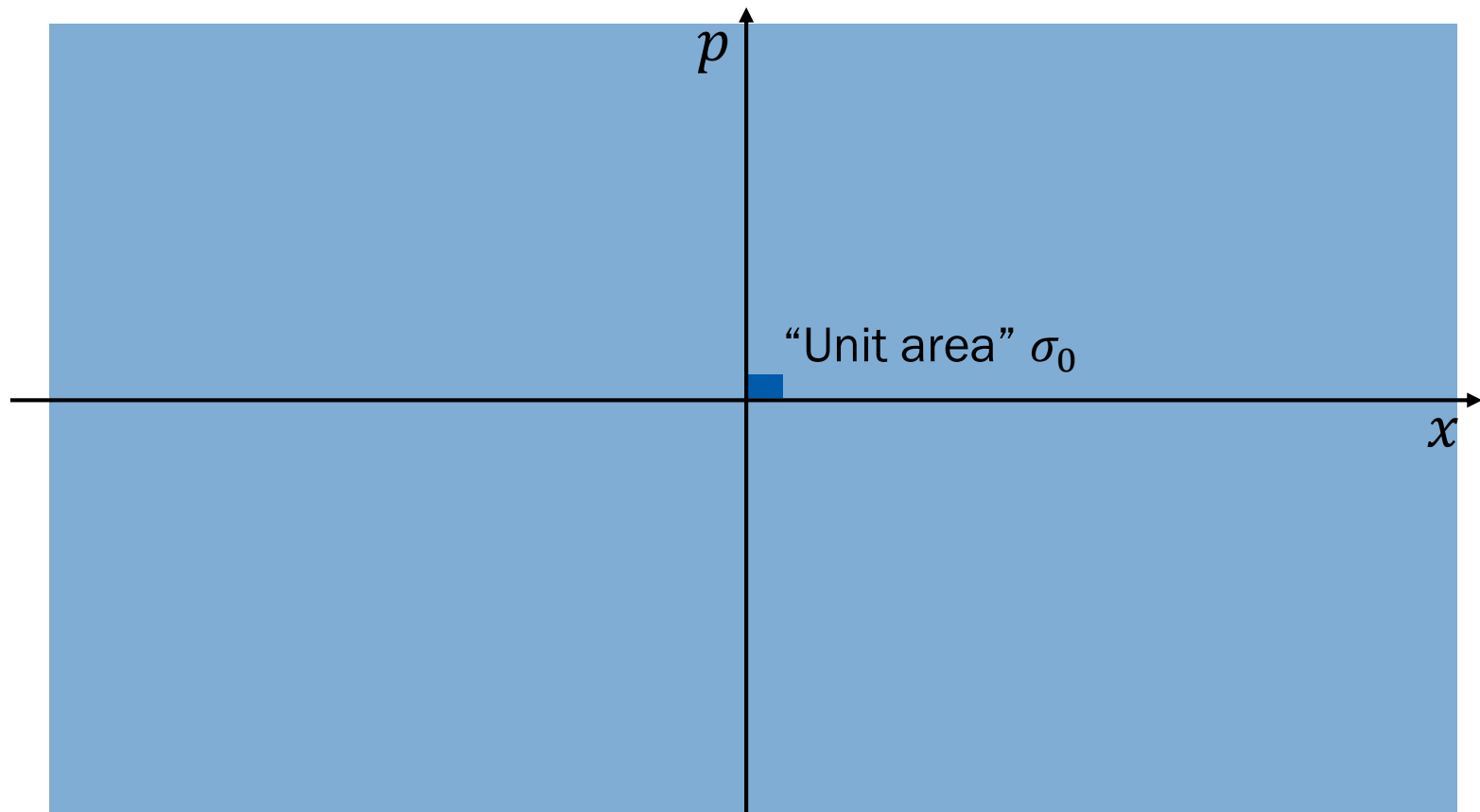
Microstate of Classical Particle



Classical Partition Function



Classical Partition Function



Classical Partition Function

$$z = \sum_s e^{-\beta E(s)} \rightarrow z = \frac{1}{\sigma_0} \int \int e^{-\beta E(x,p)} dx dp$$

for a 1-dimensional single particle

Two Classical Particles

- Consider two particles A and B, with energies ε^A and ε^B
- Partition function of A: $z_A = \frac{1}{\sigma_0} \int e^{-\beta \varepsilon^A} d^3 q_A d^3 p_A$
- Partition function of B: $z_B = \frac{1}{\sigma_0} \int e^{-\beta \varepsilon^B} d^3 q_B d^3 p_B$
- Total energy: $E = \varepsilon^A + \varepsilon^B$
- Total partition function

$$Z = \frac{1}{\sigma_0^2} \int e^{-\beta E} d^3 q_A d^3 q_B d^3 p_A d^3 p_B = z_A z_B$$

N Particles

- Distinguishable particles

$$Z = z_A z_B \cdots z_X$$

If the one-particle partition functions are identical ($= z$),

$$Z = z^N$$

- Indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

Ideal Gas (Classical Version)

The system consists of **indistinguishable particles**:

$$Z = \frac{1}{N!} z^N$$

Ideal Gas (Classical Version)

- Classical treatment of z :

$$z = \frac{1}{\sigma_0} \int \int e^{-\beta E(x,p)} dx dp$$

Since $E(x, p) = p^2/2m$,

$$z = \frac{1}{\sigma_0} \int \int e^{-\beta p^2/2m} dx dp$$

Partitioning of z

$$z = \frac{1}{\sigma_0} \int \int e^{-\beta p^2/2m} dx dp$$

$$z = \frac{1}{\sigma_0} \left[\int 1 dx \right] \left[\int e^{-\beta p^2/2m} dp \right]$$

$$\int 1 dx = \int_0^a 1 dx = a$$

Momentum Integral

$$z = \frac{a}{\sigma_0} \left[\int e^{-\beta p^2 / 2m} dp \right]$$

$$\int e^{-\beta p^2 / 2m} dp = \int_{-\infty}^{\infty} e^{-\beta p^2 / 2m} dp$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha}$,

$$z = \frac{a}{\sigma_0} \left(\frac{2\pi m}{\beta} \right)^{1/2}$$

For N Particles

$$z = a \left(\frac{2\pi m}{\beta \sigma_0^2} \right)^{1/2}$$

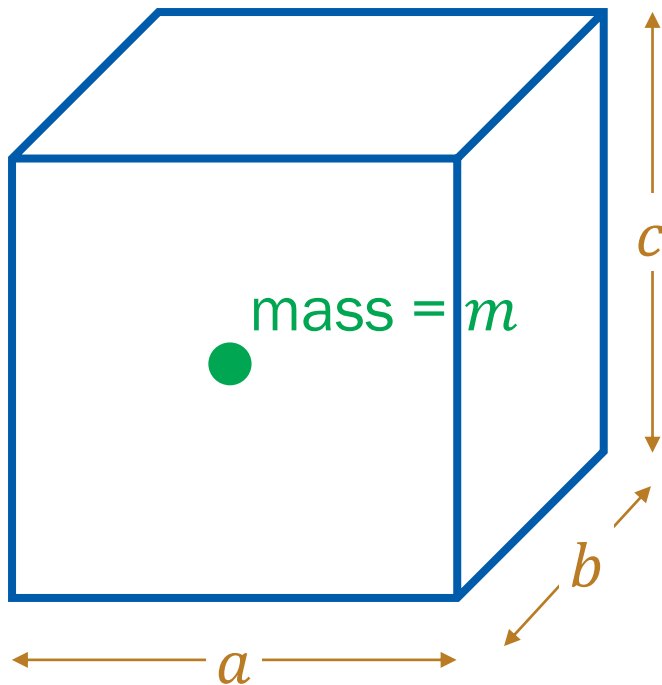
$$Z = \frac{1}{N!} \left(\frac{2\pi m}{\beta \sigma_0^2} \right)^{N/2} a^N$$

For a 3-dimensional system,

$$Z = \frac{1}{N!} \left(\frac{2\pi m}{\beta \sigma_0^2} \right)^{3N/2} a^{3N} = \frac{1}{N!} \left(\frac{2\pi m}{\beta \sigma_0^2} \right)^{3N/2} V^N$$

Particle in a 3D Box

$$\mathcal{V}(x, y, z) = \mathcal{V}_x(x)\mathcal{V}_y(y)\mathcal{V}_z(z)$$



$$\mathcal{V}_x(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \leq x \leq a \\ \infty, & x > a \end{cases}$$
$$\mathcal{V}_y(y) = \begin{cases} \infty, & y < 0 \\ 0, & 0 \leq y \leq b \\ \infty, & y > b \end{cases}$$
$$\mathcal{V}_z(z) = \begin{cases} \infty, & z < 0 \\ 0, & 0 \leq z \leq c \\ \infty, & z > c \end{cases}$$

Review: Quantum Mechanics

- Quantum numbers $n_x, n_y, n_z = 1, 2, 3, \dots$
- Energy levels

$$E = E_x + E_y + E_z = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right) = E_{(n_x, n_y, n_z)}$$

- Wavefunctions

$$\psi(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right)$$

V (volume)

Ideal Gas (Quantum Version)

The system consists of **indistinguishable particles**:

$$Z = \frac{1}{N!} z^N$$

- Quantum treatment of z :

$$E_n = \frac{h^2}{8ma^2} n^2 \quad (n = 1, 2, 3, \dots)$$

$$z = \sum_{n=1}^{\infty} e^{-\beta h^2 n^2 / 8ma^2}$$

Approximation

$$z = \sum_{n=1}^{\infty} e^{-\beta h^2 n^2 / 8ma^2}$$

Approximate

$$\sum_{n=1}^{\infty} e^{-\alpha n^2} \rightarrow \int_0^{\infty} e^{-\alpha n^2} \mathrm{d}n = \left(\frac{\pi}{4\alpha}\right)^{1/2}$$

Hence,

$$z = \left(\frac{\pi}{4\alpha}\right)^{1/2} = \left(\frac{2\pi m}{\beta h^2}\right)^{1/2} a$$

For N Particles

$$Z = \frac{1}{N!} z^N$$

$$Z = \frac{1}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{N/2} a^N$$

For a 3-dimensional system,

$$Z = \frac{1}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} a^{3N} = \frac{1}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} V^N$$

Comparison

- Classical version

$$Z = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{\sigma_0^2} \right)^{3N/2}$$

- Quantum version

$$Z = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

$$\therefore \sigma_0 = h$$

Helmholtz Energy

$$Z = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Now, we can calculate the Helmholtz energy:

$$A = -k_B T \ln Z = -k_B T \ln \left\{ \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} \right\}$$

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

Internal Energy

$$Z = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Also the internal energy:

$$U = -\frac{\partial(\ln Z)}{\partial\beta} = -\frac{\partial}{\partial\beta} \ln \left\{ \frac{V^N}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} \right\}$$

$$U = \frac{\partial}{\partial\beta} \ln \beta^{3N/2} + \frac{\partial}{\partial\beta} (\text{terms independent to } \beta) = \frac{3}{2} N k_B T$$

Pressure

$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

Using $p = -(\partial A / \partial V)_T$,

$$p = Nk_B T \frac{\partial(\ln V)}{\partial V} + \frac{\partial}{\partial V} (\text{terms independent to } V)$$

$$p = \frac{Nk_B T}{V}$$

One can obtain **entropy** similarly